

Writing Context-Free Grammars using NLTK

Instructions

You may use any programming language to do this assignment using the dataset provided here. Please submit the following:

1. A report answering the questions mentioned below
2. Your code in a zip/tarball archive

Please install the NLTK toolkit in your machine: <https://www.nltk.org/>

Introduction

This is an exercise in writing context-free grammars (CFGs) to capture generalizations about natural language syntax. You can use an automatic parser available in NLTK to check whether your grammar accounts for the key generalizations. To get you going, consider the following grammar:

```
NP → Det N
NP → NP PP
PP → P NP
Det → a
Det → an
N → joke
N → woman
N → umbrella
N → street
P → about
P → with
P → on
```

NLTK Parsing Example

Running the following code will parse the string *a joke about the woman with an umbrella on the street* with start symbol (i.e., goal category) NP, generating the five parses.

```

import nltk
from nltk import Nonterminal, nonterminals, Production, CFG

grammar = CFG.fromstring("""
NP -> Det N | NP PP
PP -> P NP
Det -> 'the' | 'a' | 'an'
N -> 'joke' | 'woman' | 'umbrella' | 'street'
P -> 'about' | 'with' | 'on'
""")

parser = nltk.parse.BottomUpChartParser(grammar)
sentence = ['a', 'joke', 'about', 'the', 'woman',
            'with', 'the', 'umbrella', 'on', 'the', 'street']

for tree in parser.parse(sentence):
    tree.pretty_print()

```

Note that NLTK's `CFG.fromstring()` function takes the left-hand-side of the first rule listed as the goal category, and allows multiple rewrites for a single category to be expressed with a disjunction on the right-hand side of a single rule, so that the rule $X \rightarrow Y_1 \dots Y_m \mid Z_1 \dots Z_n$ is shorthand for the two rules $X \rightarrow Y_1 \dots Y_m$ and $X \rightarrow Z_1 \dots Z_n$.

Questions

Part A (5 marks)

1. Visualize each tree emitted by the NLTK parser for the sentence in the above question and submit PDF output.
2. Recurse through each tree top-down fashion (starting from the root node) and print the following information: 1. Number of non-terminal nodes (nodes which aren't words) 2. Print out each word with a numerical index indicating its position in the sentence.

Part B (5 marks)

Write a context-free grammar that will capture the structural ambiguity in the sentence *They are flying planes*.

Your grammar should respect the facts that:

- (i) an NP should be substitutable with a pronoun given the right context; and
- (ii) a verb and an immediately following NP can combine to form a VP.

You can check your work with the NLTK parser to make sure that your grammar behaves the way you think it will behave.

Part C (5 marks)

Extend your grammar to capture the structural ambiguity in the sentence *Flying planes can be dangerous*.

Hint

A non-finite VP can serve as the subject of an English sentence, such as in the sentences *To err is human*. It is OK to use a unary rewrite rule with a right-hand-side element that is a phrasal category.

Part D (5 marks)

The ambiguity of the preceding sentence is eliminated if you change *can be* to either *is* or *are*. Why? Modify your grammar so that it captures this disambiguation effect.